SEYYED ARAD HOSSEINI MOGHADDAM

AI ENGINEER / DATA SCIENTIST

Medical imaging • Persian NLP • Reproducible machine learning

Website • GitHub • Hugging Face • LinkedIn

PROFESSIONAL PROFILE

I build inspectable machine-learning systems across 3D medical imaging, Persian language models, agentic retrieval, and dataset-leakage auditing. My developing research direction connects reproducible AI engineering with psychology and neuroscience.

SELECTED TECHNICAL WORK

3D cardiac MRI segmentation PYTORCH / MONAI

Implemented a 3D U-Net pipeline for left-atrium segmentation on MSD Task02 Heart, covering NIfTI preprocessing, patch training and sliding-window inference. Project records report Dice 0.8821; this result is author-reported, not independently or clinically validated.

MRI project /

CohortSeal + Holdout Ledger EVALUATION INTEGRITY

Built open research-software tools to audit dataset splits for group leakage and exact duplicates. Documented testable, reviewable integrity checks rather than claiming clinical verification.

CohortSeal / Holdout Ledger /

Persian language-model research TRANSFORMERS / NLP

Released public Persian language-model and tokenizer artifacts. Audited exported checkpoint metadata: 134.99M stored tensor elements versus a 110.4M architecture claim; runtime weight tying and reported language-model metrics remain unverified.

Persian GPT / Model audit /

AI Research Assistant Platform LANGGRAPH / FASTAPI

Developed agent-based retrieval and research workflows with structured synthesis, API integration, and offline tests. Published tests do not imply externally verified answer quality.

Research assistant /

Persian NLP + psychology-language tools QLORA / TRANSLATION

Published experimental English–Persian translation and text-to-JSON models, plus a psychology-terminology desktop workbench. Models are not clinically validated.

PsyLex Studio /

TECHNICAL TOOLKIT

Python / PyTorch / MONAI / Transformers / Hugging Face / LoRA & QLoRA / LangGraph / RAG / FastAPI / GitHub Actions / Docker / statistical analysis

RESEARCH DIRECTION

Computational psychology and neuroscience, transparent benchmark design, medical AI evaluation, data provenance, and reproducible scientific software. Interested in well-scoped research collaborations.

Evidence note Project metrics are author-reported unless explicitly audited.
No degrees, research affiliations, clinical credentials, or publications are asserted.